Neon Haven: Technical Sheet

Summary

Single self-contained HTML file (~35 KB), with no libraries, no images, no network requests and no external files. It runs offline in any modern browser and is built mobile first.

I used plain HTML5 Canvas 2D and vanilla JavaScript rather than Phaser, Three.js or React, so it loads instantly with nothing to download.

Technology used

Rendering: Canvas 2D isometric projection on an 18×18 grid. Buildings are procedurally drawn 3-face prisms with neon edge glow (shadowBlur), animated window lights, beacons, smoke, rain and lightning flashes. Additive-blend particles, a radial ground glow, twinkling stars, animated water and a CSS vignette. Rendering is DPR-aware (up to 2×).

UI: HTML/CSS overlay with glassmorphism (backdrop-filter), gradient text with an animated glow, CSS transitions between screens, an animated banner and safe-area insets for notched phones.

Audio: Web Audio API only. The SFX are synthesised from oscillators and filtered noise (place, road, coin, upgrade, error, fire, explosion, milestone, defeat). Music is a generative synthwave loop (pads, bass, arpeggio, hats through a feedback delay) whose tempo rises with the year and whose chords darken when the city is stressed. A compressor sits on the master bus.

Input: Pointer Events with one-finger tap and drag, two-finger pinch zoom and mouse-wheel zoom. Taps and drags are told apart by an 8-pixel threshold.

Saving: localStorage. The city auto-saves every 3 months, on menu and when the tab is hidden. Legacy points, upgrades and the top 10 high scores are saved separately.

Game systems

Monthly tick (4 s at 1×, plus 2× and 4×). Each month the game recomputes road access, power coverage (Manhattan radius) with brownout scaling, population, jobs, food, happiness (parks, pollution, clinics, unemployment, hunger, blackout, difficulty) and income (wages + taxes × happiness × perks – upkeep scaled by difficulty).

Difficulty: diff = 1 + 0.18 × year. It raises upkeep and the happiness penalty, and disaster chance is 3% + 1.5% per year. Storm damage and earthquake count scale with the year. Build prices rise 2% a year.

Juice: screenshake scaled to the event, floating money/damage/level numbers, burst particles, a pop-in bounce animation on build, milestone banners at 50/100/250/500/1000 pop.

Self-audit 1: balance (automated)

I wrote a headless simulator that runs the real game logic in Node with a rule-based bot player. On the first version, all 20 runs went bankrupt and peak population got stuck at 6 to 36. Wages were too low and starting happiness was below 40.

Fixes: wages from 2.2 to 4 per worker, tax ×1.6, upkeep cut by a third, base happiness from 55 to 64, starting money from $1,500 to $2,000, a softer early storm, and upkeep that scales with difficulty so late-game cash can't pile up forever.

Result over 20 runs: the average city lasts about 21 years, and 3 in 20 survive the full 30-year test. The difficulty curve now builds up instead of killing cities early.

Self-audit 2: runtime and UI (automated)

I ran the full game, UI and renderer in Node with a stubbed DOM and canvas: menu, building, upgrade, demolish, 65 months of play, fire, shop, save, menu, continue, and 400 animation frames.

Bug found and fixed: on a browser without Web Audio the game crashed on the first tap. Audio is now optional and wrapped safely, and it resumes suspended audio on iOS.

I confirmed that fire spreads and destroys unprotected buildings, unpowered and roadless buildings switch off and show the right icon, and saving works.

Self-audit 3: player experience

I added a fire warning to the tip system (it had stayed silent while a city burned), ordered the tips by urgency (start, fire, blackout, hunger, jobless, empty jobs, unhappy, losing money), and added the "debt x/3" countdown on the date so a loss never comes as a surprise.

Still open: I tested it in simulation and with a stubbed browser, not on a real phone, so please try it on yours. Next upgrades worth considering: a building info panel on long-press, road drag-painting, zones, day/night lighting and more disaster types.

Files and storage

Neon-Haven.html (the game), Neon-Haven-How-to-Play.pdf, Neon-Haven-Technical-Sheet.pdf. Save keys: neonhaven_v1_save and neonhaven_v1_meta.